# SOCIAL NETWORK ANALYZER

## 1. ABSTRACT

This project focuses on the development of a Social NetworkAnalyzer using Python and SQL to visualize and analyze connections between entities. The system leverages advanced data structures like deque for efficient traversal and NetworkX (nx) for graph-based relationship modeling.

The application allows users to input entities and define relationships, which are then represented as a graph. It helps in identifying connections, shortest paths, and relationship patterns. Colorama is used to enhance command-line output for better readability.

This project demonstrates how graph theory and database systems can be combined to solve real-world relationship mapping problems.

## 2. INTRODUCTION

In many real-world scenarios such as social networks, organizational structures, and recommendation systems, understanding relationships between entities is crucial.

This project implements a command-line-based Relationship Mapper System where users can create, explore, and analyze connections between different entities. Python is used for implementing logic, SQL for structured data storage, NetworkX for graph operations, deque for efficient traversal, and Colorama for enhanced CLI output.

The system simulates real-world graph-based systems used in platforms like LinkedIn, Facebook, and knowledge graphs.

## 3. OBJECTIVES

- To develop a system for mapping relationships between entities
- To implement graph-based data structures using NetworkX
- To efficiently traverse relationships using deque
- To store and manage data using SQL
- To visualize and analyze connections between nodes
- To enhance CLI experience using Colorama
- To demonstrate real-world applications of graph theory

## 4. SYSTEM MODULES

### *4.1 User Module*

- Add entities (nodes)
- Define relationships (edges)
- View connections
- Search relationships

### *4.2 Graph Module (NetworkX)*

- Create and manage graph structures
- Add/remove nodes and edges
- Perform graph analysis (degree, connectivity, etc.)

### *4.3 Traversal Module (Deque)*

- Implement BFS (Breadth-First Search)
- Efficient queue operations
- Find shortest paths between entities

### *4.4 Database Module (SQL)*

- Store entities and relationships
- Maintain structured data
- Perform CRUD operations

### *4.5 Visualization/Output Module (Colorama)*

- Highlight outputs with colors
- Improve readability of CLI results
- Differentiate nodes, paths, and alerts

## 5. FEATURES OF THE SYSTEM

- Graph-based relationship mapping
- Efficient traversal using deque
- Shortest path detection between nodes
- Colored CLI output using Colorama
- Persistent storage using SQL database
- Modular and scalable architecture
- Real-time relationship analysis

## 6. USAGE OF THE SYSTEM

The system operates through a command-line interface.

Users can:

- Add entities (e.g., people, organizations, items)
- Define relationships between them
- View the entire network structure
- Find connections between two entities
- Analyze shortest paths using BFS

The system processes relationships as a graph and provides meaningful insights such as direct and indirect connections.

## 7. DATABASE TABLES

The system uses a relational database with the following tables:

*Entities Table*

Stores all nodes:

- Entity ID
- Entity Name
- Type (optional)

*Relationships Table*

Stores connections:

- Relationship ID
- Source Entity ID (Foreign Key)
- Target Entity ID (Foreign Key)
- Relationship Type

*Logs Table (Optional)*

Stores activity logs:

- Log ID
- Action performed
- Timestamp

Indexes are used to optimize search and retrieval performance.

## 8. ADVANTAGES

- Efficient graph-based modeling
- Fast traversal using deque
- Scalable for large datasets
- Clear visualization in CLI using Colorama
- Real-world applicability (social networks, recommendation systems)
- Modular and extendable design

## 9. LIMITATIONS

- No graphical user interface (GUI)
- Limited visualization (CLI-based only)
- Requires basic understanding of graphs
- Not optimized for extremely large-scale graphs
- No real-time web integration

## 10. FUTURE ENHANCEMENTS

- Develop a web-based interface with graph visualization
- Integrate interactive graph libraries (e.g., D3.js)
- Add AI-based relationship predictions
- Implement weighted graphs and advanced algorithms
- Enable real-time collaboration
- Add export features (PDF/Graph images)

## 11. CONCLUSION

The Relationship Mapper System successfully demonstrates how graph theory concepts can be implemented using Python and SQL. By integrating NetworkX for graph operations, deque for efficient traversal, and Colorama for enhanced output, the system provides a powerful tool for analyzing relationships.

Although currently CLI-based, the project lays a strong foundation for building advanced systems such as social network analyzers, recommendation engines, and knowledge graphs.

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